



Parametric Aerodynamic Study of the Properties of Leading-Edge Serrations on Wing Sections

Jordan J. Whittaker* and Con Doolan†
University of New South Wales, Sydney, NSW, 2052, Australia

In this paper, the aerodynamics of various geometric properties of leading-edge serrations on wing sections are explored through automated large-eddy simulation (LES) computational optimisation in ANSYS Fluent, with experimental verification. The wavelength, amplitude, and termination distance of the serrations are parameterised on the cambered NACA2412 airfoil for an aspect-ratio two wing-section. For a Reynolds number of 100,000, the pre-stall, stall, and post-stall regimes are explored, and the relative aerodynamic performance of the serration combinations is found. The study found that in the pre-stall regime, smaller, less perturbing serrations are favoured for their reduced drag, with a slight but unanimous performance reduction compared to the unserrated wing section. As the wing-sections are pushed to stall and post-stall regimes, the opposite is found. The large-amplitude serrations dominate the post-stall regime, which correlates with their ability to create stable vortex structures that persist beyond 50% of the chord. The serration combination that maintains performance across all three regimes has a wavelength of 1/6.5, an amplitude of 0.03, and a termination distance of 0.35 (specific to the average chord). A strong correlation between the wavelength-to-amplitude ratio and optimal configurations is found, with optimal configurations occurring at a ratio of 5.5.

I. Nomenclature

A	=	amplitude of serrations
$\overline{C}$	=	average chord of wing-section
C_L, C_D	=	lift and drag coefficients
C_{L_0}, C_{D_0}	=	lift and drag coefficients of unserrated wing-section
Re	=	Reynolds Number
S	=	span of wing-section
T	=	termination distance of serrations
W	=	wavelength of serrations
w, a, t	=	average chord specific serration properties

II. Introduction

LEADING-EDGE (LE) serrations are biomimetic lift-enhancing devices that are hypothesised to improve the aerodynamic performance of wings through the creation of stream-wise vortex pairs that re-energise the boundary layer and delay flow separation. Quasi-sinusoidal LE serrations are found on the pectoral flippers of Humpback whales (*Megaptera novaeangliae*) seen in Fig. 1 and hypothesised to contribute to their high hydrodynamic manoeuvrability despite their large size and rigid bodies. They were first investigated by Fish and Battle (1995) [1], who observed an increase in lift, especially in the high-angle-of-attack (AoA) regime, a delay in stall, and a reduction in drag. The most common results of the LE-serrations in literature are a softening of the stall profile, replacing the sudden stall onset with a parabolic lift peak best observed in Hansen (2012) [2] or a partial stall onset replacement observed in Johari et al. (2007) [3] with a significant performance improvement post-stall. Immediately post-stall of an unserrated airfoil is where the most improvement is seen with up to a 40% [1–3] increase in lift. There are notable studies that explore variations in serration properties, such as Lacagnina (2021) [4] and Fernandes et al. (2013) [5]. However, the effect of the variation of serration properties remains unknown. A study by Stanlov and Chong (2019) [6] suggested that the optimal serration amplitude occurs at 2.5% of chord and a wavelength of between 1/10 and 1/5 of chord in their six serration combination study.

*Undergraduate, Mechanical and Manufacturing Engineering, jordan.whittaker@student.unsw.edu.au. AIAA Student Member 1863689

†Professor, Mechanical and Manufacturing Engineering, AIAA Associate Fellow



Fig. 1 Sinusoidal LE tubercles of a humpback whale's (*Megaptera novaeangliae*) flipper.

This paper aims to determine how variations in the geometric properties of LE-serrations affect the aerodynamic performance of wings. The wavelength, W , the amplitude, A , and the termination distance, T , of the serrations are parameterised. A computational fluid dynamics (CFD) automation workflow in ANSYS Fluent, executed on high-performance computing clusters, is set up to simulate parameter combinations and undergo experimental validation on the University of New South Wales' (UNSW) Educational Wind Tunnel (EWT).

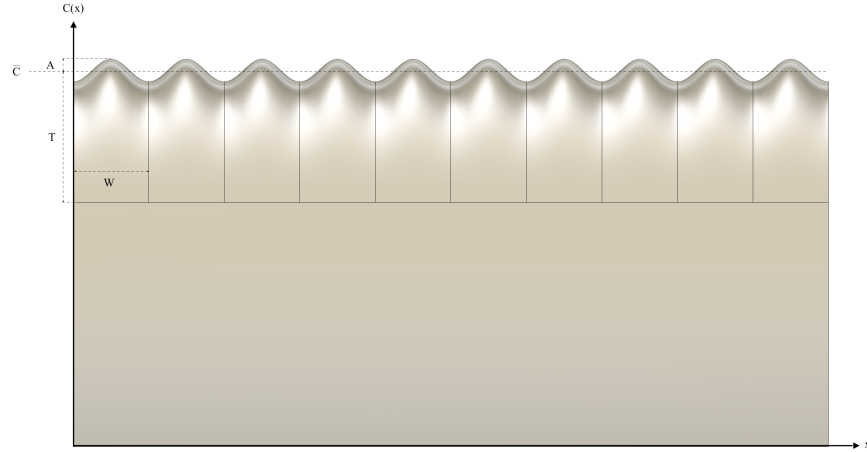


Fig. 2 Parameter dimensions of the sinusoidal LE serrated airfoil including amplitude, A , wavelength, W , termination distance, T , mean chord, $\bar{C}$, and variable chord profile, $C(x)$, following Eq. 1.

III. Methodology

A. Serrated Wing Section

The sinusoidal serrated wing sections were designed and parametrised in Fusion 360 with variable control over the wing properties of average chord length, $\bar{C}$, and span, S , and serration properties of wavelength, W , amplitude, A , and termination distance, T . All properties are taken specific to the average chord length and are represented as w , a , and t . The variable serration chord profile follows the function;

$$C(x) = \bar{C} - A \cos \frac{2\pi x}{W}, \quad (1)$$

With the termination distance, T , defining the section of the airfoil from the LE that is scaled to match the chord profile defined in Eq. 1 (e.g. at a value of $t = 1$, the entire airfoil profile is scaled to $C(x)$). These profiles ensure that

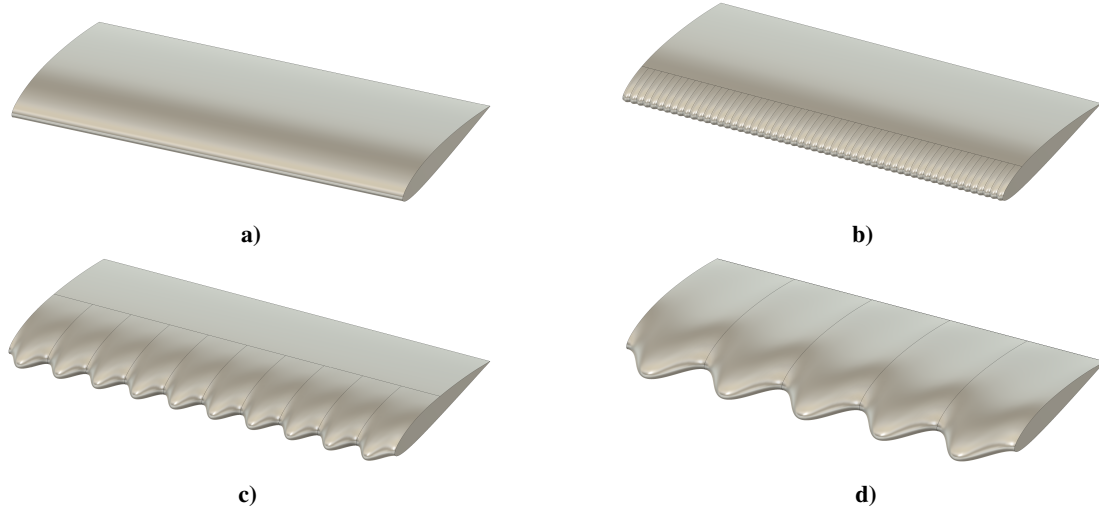


Fig. 3 Models with the serration combinations of a) unserrated, b) $w = 1/28.5$, $a = 0.005$, $t = 0.2$, c) $w = 1/5$, $a = 0.05$, $t = 0.5$, d) $w = 1/2.5$, $a = 0.1$, $t = 1.0$.

the plan-form area used to calculate aerodynamic parameters remains constant across all models. The wavelengths chosen to be analysed are all integer values of the span to ensure that the wing-section is always symmetric about the centre chord, where the line of symmetry either falls on a trough or a crest. These properties are dimensioned in Fig. 2. The average chord of the model was taken as $\bar{C} = 100$ mm with an aspect-ratio (AR) of two and thus, span, $S = 200$ mm. $AR = 2$ is chosen so that the finer serration combinations can be 3D-printed, surface conditioning is maintained, and the study remains as general as possible. A range of parameter combinations can be seen in Fig. 3. The chosen airfoil profile is the NACA2412 for the broadest application of this study, following from Hansen (2012) [2] who investigated the effectiveness of LE-serrations on several cambered airfoils. The full geometric serration properties explored in this report are;

$$w = W/\bar{C} = \left[\frac{1}{50}, \frac{1}{28.5}, \frac{1}{20}, \frac{1}{13.5}, \frac{1}{10}, \frac{1}{6.5}, \frac{1}{5}, \frac{1}{4}, \frac{1}{3.5}, \frac{1}{3}, \frac{1}{2.5}, \frac{1}{2} \right], \quad (2)$$

$$a = A/\bar{C} = [0.0025, 0.005, 0.01, 0.015, 0.02, 0.025, 0.03, 0.04, 0.05, 0.075, 0.1, 0.15], \quad (3)$$

$$t = T/\bar{C} = [0.1, 0.2, 0.35, 0.5, 0.75, 1.0]. \quad (4)$$

Only the wavelength and amplitude values will be explored entirely at a single initial value of termination at 0.35, which is the most commonly explored termination distance in the literature, second only to fully inserted serrations. Further termination distances will be explored to a lesser extent. Some serration combinations are unrealistic and have been omitted from the study. Following Joseph and Sathyabhama (2022) [7] investigation into variable effect of Re on serration performance, a value of $Re = 1 \times 10^5$ will be explored.

B. Computational Method

To reduce the amount of manual simulation setup the study needed, the CFD was set up to run automatically on two available computational clusters, namely Katana at UNSW, a 7,600 core cluster and GADI at the National Computational Infrastructure (NCI), a 260,000 core cluster, both of which were given access to by ResTech Services at UNSW. A BASE_SIM was set up (a third of the total computational work was devoted to refining the base setup), designed to be optimal across the entire parameter space. The necessary manual work is selecting the named selections to define local surface-mesh refinement and lift/drag calculation regions. To submit a job, the named selections are completed, and the following job parameters are defined in a Python script: model, #CPU, memory, walltime, AoA range, velocity range, and cluster. The script then copies the BASE_SIM across the entire AoA and velocity sweep, uploads the files to the specific cluster and submits each simulation per model as an array job. The lift and drag profiles are automatically exported at completion of the job. A separate script scrapes the exported data locally and processes it.

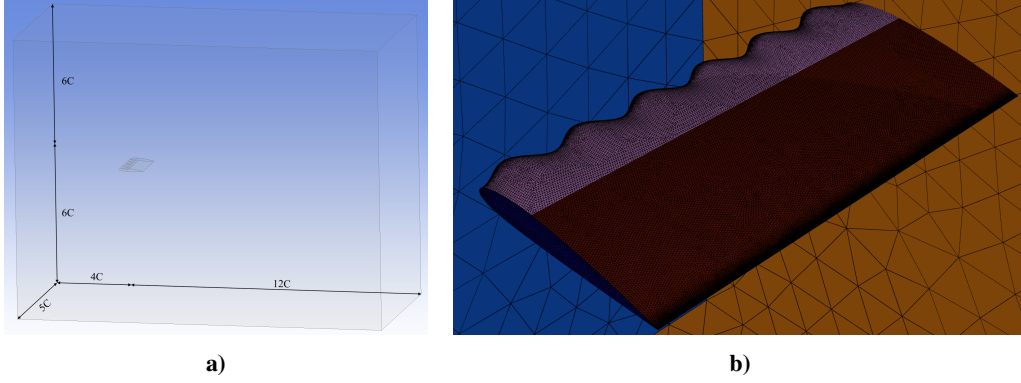


Fig. 4 Figures of the a) dimensioned fluid domain, and b) surface mesh fidelity.

Through multiple convergence studies, the fluid domain and mesh were determined. Whereas the goal of the study was to be as accurate and generalised as possible, in order for the results to be experimentally verifiable, the fluid domain was not taken completely to null delta convergence and was kept closer to the dimensions of the wind-tunnel span-wise. This is also why periodic boundary conditions were omitted, and a standard fluid domain was chosen. The dimensions of the fluid domain, seen in Fig. 4a), are $\pm 5\bar{C}$ vertically, $12\bar{C}$ downstream, $4\bar{C}$ upstream, and $\pm 2.5\bar{C}$ span-wise. These values are defined from the centre of the quarter chord of the wing section. Surface mesh convergence was achieved at approximately 15 cells per wavelength at the smallest wavelength, yielding surface meshes with 200,000 to 600,000 faces. Mesh fidelity is shown in Fig. 4b). Whereas $k - \omega$ SST models have been proven effective at predicting the performance of LE-serrations seen in Ali et al. (2024) [8], Large Eddy Simulations (LES) will be used for this study as it best models the hypothesised stream-wise vortices LE-serrations generate (a considerable factor for the LES model is the computational availability of both clusters). As LES requires full viscous-sublayer resolution, a 30-layer boundary layer across the entire wing surface was added with $y^+ \sim 1$. The wake region was refined for a cell size of $\sim 0.01\bar{C}$ to capture vortex generation. The final volume cell number ranged from 4,000,000 to 15,000,000 cells. To reduce computational work, the case is initially solved with a time-step equivalent Courant–Friedrichs–Lewy (CFL) number of 10 – 50 for the first 200 time-steps to push the model past stall and is then reduced to a CFL of 1 – 5 for an additional 1000 time-steps to ensure vortex generation resolution. Final values were taken as averages of the final 600 time-steps or through a post-convergence criterion. These simulations ranged from 4 to 12 hours per solution to run on 64-core nodes. In total, 502 simulations were run for data collection, and the total computational work for both collected data and R&D was 406,500 core-hours across both clusters. As the settings for each simulation are optimised to be accurate across a range of models rather than specific to a single model, the results in this report are presented as trends rather than explicitly accurate values.

C. Experimental Method

The experimental setup was simple and was intended only to verify the accuracy of the trends in the computational results. The dimensions of the EWT seen in Fig. 5 are 30.48 cm $\times$ 30.48 cm $\times$ 60.96 cm equipped with a 3-component sting balance with max AoA of 23° . As the simulated domain is greater than this value, a mount was designed to offset the model by 30 mm (this also reduces sting balance interference with the flow), which allowed for a -5° offset to be designed into the mount. This expanded the maximum AoA to 28° , which sufficiently contains the computational domain. The models were 3D-printed from PLA, with the print "grain" constant in the stream-wise direction across all models. The AoA step was large at the edges of the sweep, with $\Delta 0.5^\circ$ steps over the centre of the sweep at stall.

IV. Results and Discussion

A. Experimental Verification

To verify the accuracy of the computational setup and to determine the necessary AoAs to simulate, a larger AoA sweep was conducted experimentally and computationally across an unserrated and a low- and high-fidelity serration combination. The results are compared in Fig. 6.



Fig. 5 Experimental wind-tunnel setup with stand-off mount in the EWT.

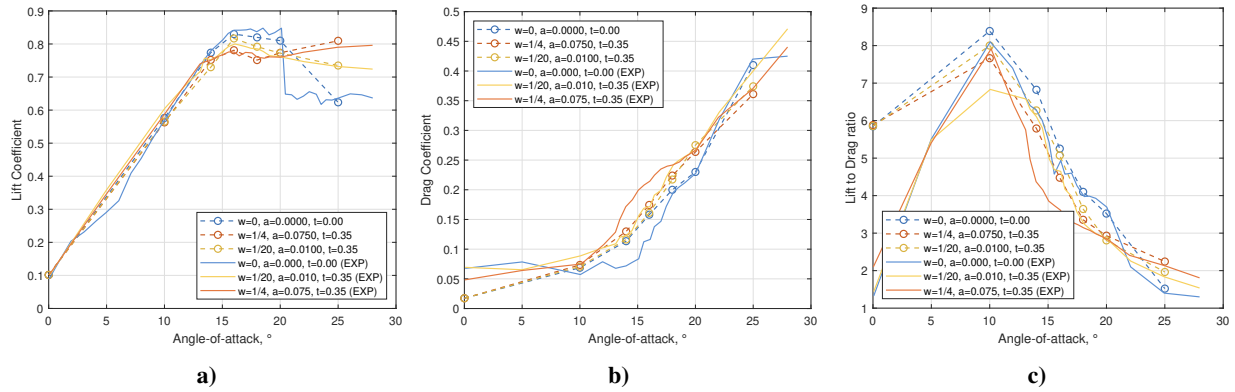


Fig. 6 Stall profiles comparing experimental data and computational data for three models across the aerodynamic properties a) Lift Coefficient, b) Drag Coefficient, and c) Lift to Drag ratio.

The lift profiles in Fig. 6a) captured by both the computational and experimental measurements are highly comparable, with nearly all phenomenological differences in model profiles captured for both. Whereas the exact stall angle is not determined computationally, the values for the coefficient of lift deviate a maximum of 5% for any of the models. This result is surprisingly accurate. The differences increase with the values for drag in Fig. 6b). The drag across all models is under-estimated by over 50% at AoA = 0°. However, for both serrated airfoils, the values return to the experimental trends, with mild deviations in the lower-fidelity serration around 15°. Past this value, the drag values for both the experimental and the computational results closely agree in the post-stall regime. The unserrated airfoil computational values deviate significantly until approximately 16°. This pre-stall inaccuracy is surprising and is best seen in Fig. 6c), where the computational values overestimate the experimental data by up to 6 times. This is most likely due to the model's surface roughness, which is not computationally simulated. For the sake of accuracy, the pre- 15° AoA regime will not be used in the parametric optimisation, as accuracy cannot be guaranteed. Otherwise, the closeness of both the computational and experimental data seen past 15° gives confidence to the accuracy of a parametric optimisation for AoA = 16°, 20°, and 25° which represent the pre-stall, stall, and post-stall performance of the serrated wing-sections respectively and will give enough fidelity to see performance trends in all three regimes.

B. Parametric Optimisation

1. Wavelength and Amplitude

The value of termination distance, t , will be held constant at the value 0.35 for the initial optimisation, and the variable effect of termination distance will be explored in a later section of this report. For succinct analysis, a specific serration combination (SC) will be referenced in the format of $[w, a, t]_{SC}$ in this section. As stated in Section. IV.A, the parametric optimisation will be run across AoA = 16°, 20°, and 25° which represent the performance of the wing-sections in the pre-stall, stall, and post-stall regimes respectively. Seen in Fig. 7, the serration intensity increases

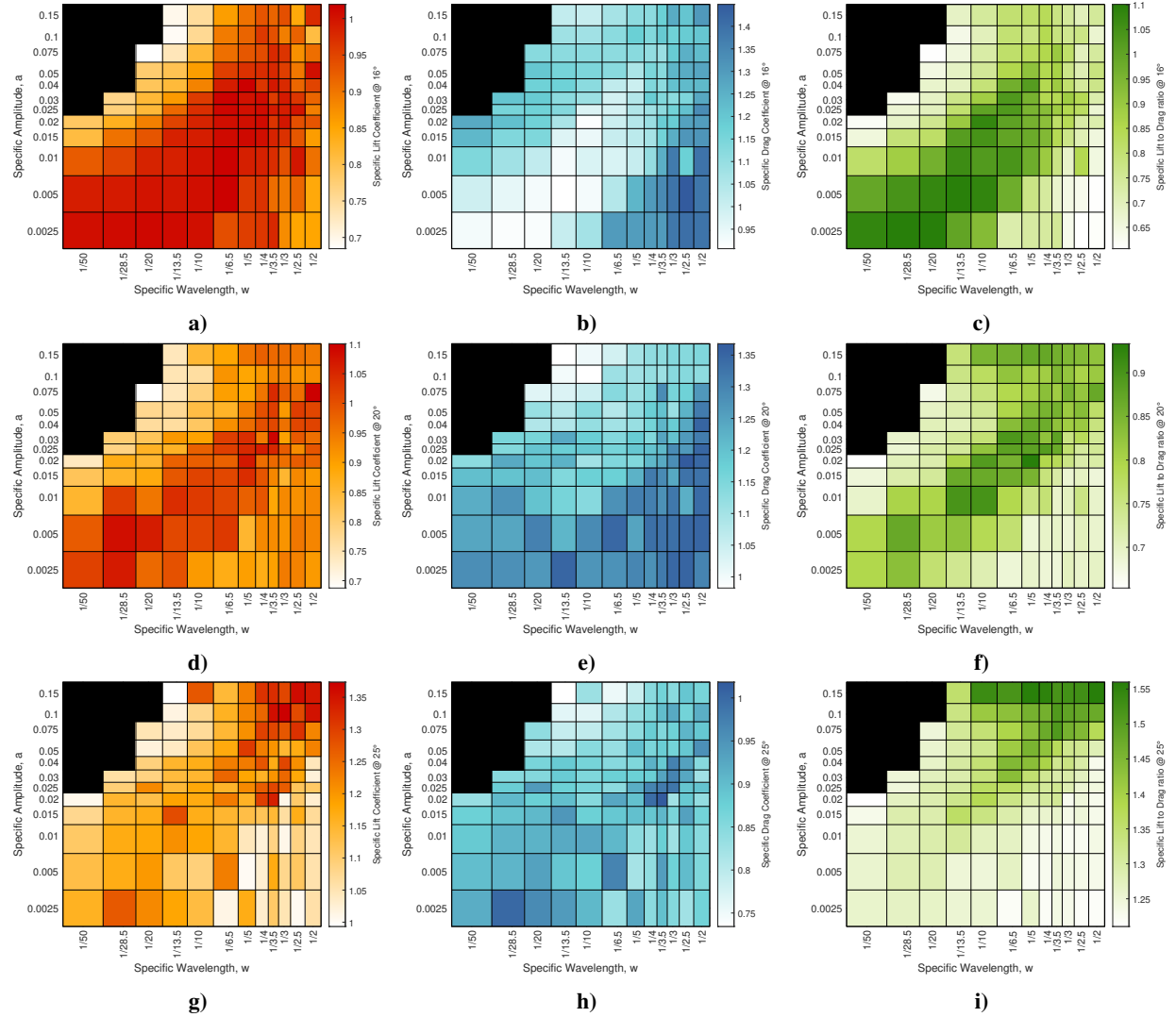


Fig. 7 Log-log plots of the parameter combinations of specific wavelength, w , and specific amplitude, a . Pre-stall regime of $\text{AoA} = 16^\circ$ for the aerodynamic properties a) Specific Lift Coefficient for $C_{L_0} = 0.83$, b) Specific Drag Coefficient for $C_{D_0} = 0.16$, and c) Specific Lift to Drag ratio for $C_{L_0}/C_{D_0} = 5.20$. Stall regime of $\text{AoA} = 20^\circ$ for the aerodynamic properties d) Specific Lift Coefficient for $C_{L_0} = 0.81$, e) Specific Drag Coefficient for $C_{D_0} = 0.23$, and f) Specific Lift to Drag ratio for $C_{L_0}/C_{D_0} = 3.52$. Post-stall regime of $\text{AoA} = 25^\circ$ for the aerodynamic properties g) Specific Lift Coefficient for $C_{L_0} = 0.62$, h) Specific Drag Coefficient for $C_{D_0} = 0.41$, and i) Specific Lift to Drag ratio for $C_{L_0}/C_{D_0} = 1.51$.

as you move from the bottom right-hand to the top-left hand corner of the plots following the increase in the parameter w/a . The black sections of these plots are the serration combinations deemed unrealistic and not simulated (eliminated from the solution space based on inspection, significant reductions in aerodynamic trends, and mesh size/quality).

The pre-stall trends at 16° observed in Fig. 7a), 7b), and 7c) are extremely clear with a wide "island" of well performing serration combinations following the line between $[1/50, 0.0025, 0.35]_{SC}$ and $[1/4, 0.04, 0.35]_{SC}$ with the approximate gradient w/a falling between 4 to 12. This result is loosely corroborated by Lacagnina et al. (2021) [4] who found the optimal w/a ratio to be between 4 and 6 in their experimental parametric study. Seen in Fig. 7a), the high-lift "island" is wide and contains the majority of the serration combinations, which indicates that pre-stall lift is relatively insensitive to serration geometry, given that w/a does not fall far from the aforementioned values of w/a and the serration amplitude is not too large. The dominance of performance in the low-amplitude range indicates that, in pre-stall regimes, small-amplitude and wavelength combinations are favoured. The observed "island" of low drag

serration combinations in Fig. 7b) is noticeably thinner than that of lift. This stark increase in drag across very-low- and very-high-fidelity serrations is best seen in Fig. 7c). The increased drag at low-fidelity serrations dominates the C_L/C_D values, significantly reducing the performance of the wing sections. This suggests that pre-stall performance is determined by the serrations' ability to reduce drag and to minimally perturb the wing's flow, not by their lift-enhancing ability. The majority of the aerodynamic values for serration combinations do not exceed those of the unserrated wing-section pre-stall. This follows the trends seen in previous literature, which state that LE-serrations are stall- and post-stall performance-enhancing devices. The least perturbing serrations at the bottom left-hand corner of the Fig. 7a) interestingly perform the worst of the entire parameter space, indicating that perturbing the flow without significant serration size seems to break the attached flow seen in the unserrated wing-section without introducing the benefits seen in the other regimes.

The parametric optimisation at stall of 20° is found in Fig. 7d), 7e), and 7f). As seen in the experimental validation and following trends of previous literature, LE-serrations mitigate sudden stall-onset. The 20° values are indicative of the performance of the stall-transient region observed in Fig. 6, where the unserrated wing-section's sudden stall onset would occur. The data for 20° are less clear for both lift and drag than those for pre-stall. The lift values seen in Fig. 7d) follow similar trends to those observed pre-stall but with a significant reduction in "island" width, indicating the performance of LE-serrations around stall is property dependent. The performance trend in lift is still following the pre-stall trends with a performance increase for an estimated gradient w/a between 5 and 7.5 starting at $[1/50, 0.0025, 0.35]_{SC}$. The difference is that larger amplitudes are favoured around stall, indicating that stall-flow is sensitive to serration amplitude. Lift performance is maintained through all wavelengths with the trend terminating at $[1/2, 0.075, 0.35]_{SC}$. The stall drag values seen in 7e) favour the high-amplitude serrations significantly, with a significant increase in drag unanimously for all $a < 0.01$ and most low-amplitude, low-wavelength serration combinations seen at the bottom right of the plot. This results in a reduction in C_L/C_D seen in Fig. 7f) of low-amplitude serrations dominated by the increase in drag. The C_L/C_D has a distinct "island" of increased performance surrounding $[1/6.5, 0.02, 0.35]_{SC}$ which lightly extends down to the extremum following the line of constant $w/a \approx 5.5$. This suggests a strong correlation between serration properties and stall performance, tightening the high-performing combinations relative to pre-stall and shifting trends towards higher-amplitude, more flow-perturbing serrations. No serration combination's performance exceeds that of the unserrated wing section at stall. The smallest reduction in lift on drag ratio is 5.2% for $[1/6.5, 0.03, 0.35]_{SC}$. This follows trends reported in previous literature, which indicate that LE-serrations mitigate the sudden onset of stall but reduce $C_{L_{max}}$ minimally [2].

The post-stall regime of 25° seen in Fig. 7g), 7h), and 7i) significantly tightens the "island" of high-performance serration combinations, with favouring of the largest serration amplitudes that are simulated in this study. Whereas the trends in lift and drag performance seen in Fig. 7g) and 7h), respectively, are much noisier than pre-stall and stall parameter optimisation data, the relative performance of the serrations post-stall seen in the C_L/C_D values in Fig. 7i) are clear. Almost all low-fidelity serration combinations hover around a 30% increase in lift-to-drag ratio post-stall, with the majority of high-performing wing sections having medium-low wavelengths and high amplitudes. This indicates a strong correlation between serration properties and performance, suggesting that the majority of the effect of creating stream-wise vortices occurs post-stall. Despite the majority of high-performing serration combinations falling at high amplitudes, all serrated combinations studied exceed the performance of the unserrated airfoil with a maximum performance increase in lift to drag ratio of 55% for $[1/5, 0.15, 0.35]_{SC}$. The most interesting result seen in the post-stall regime is the "tail" of high-performing serration combinations that falls down the previously indicated line of $w/a \approx 5.5$ from the "island" seen in the top right of the plot, which ends at $[1/6.5, 0.03, 0.35]_{SC}$.

The main result from the parametric optimisations is that the performance of LE-serrated wings is highly dependent on the regime in which the wing section is placed. For most applications, the slight reduction in performance pre-stall and at stall is enough to deny the use of LE-serrations on wings. However, for aircraft that nominally operate at high AoAs, their use could be justified. To mitigate pre-stall and stall performance reductions, the most obvious result of this optimisation is to choose the serration combination that minimises pre-stall performance reduction while maintaining post-stall vortex generation and flow separation delay. The serration combination observed at the termination of the "tail" in Fig. 7i), $[1/6.5, 0.03, 0.35]_{SC}$, is the most promising specific serration combination as it is contained in all three high-performing "island" regions. For aircraft that wish to explore the use of LE-serrations, the recommendation is to start at the indicated optimal configuration and work up or down the line of constant $w/a \approx 5.5$, depending on the specific use case. The results from this section are promising for the future usage of LE-serrations and are hopefully a fantastic starting point for a specific case investigation in industry.

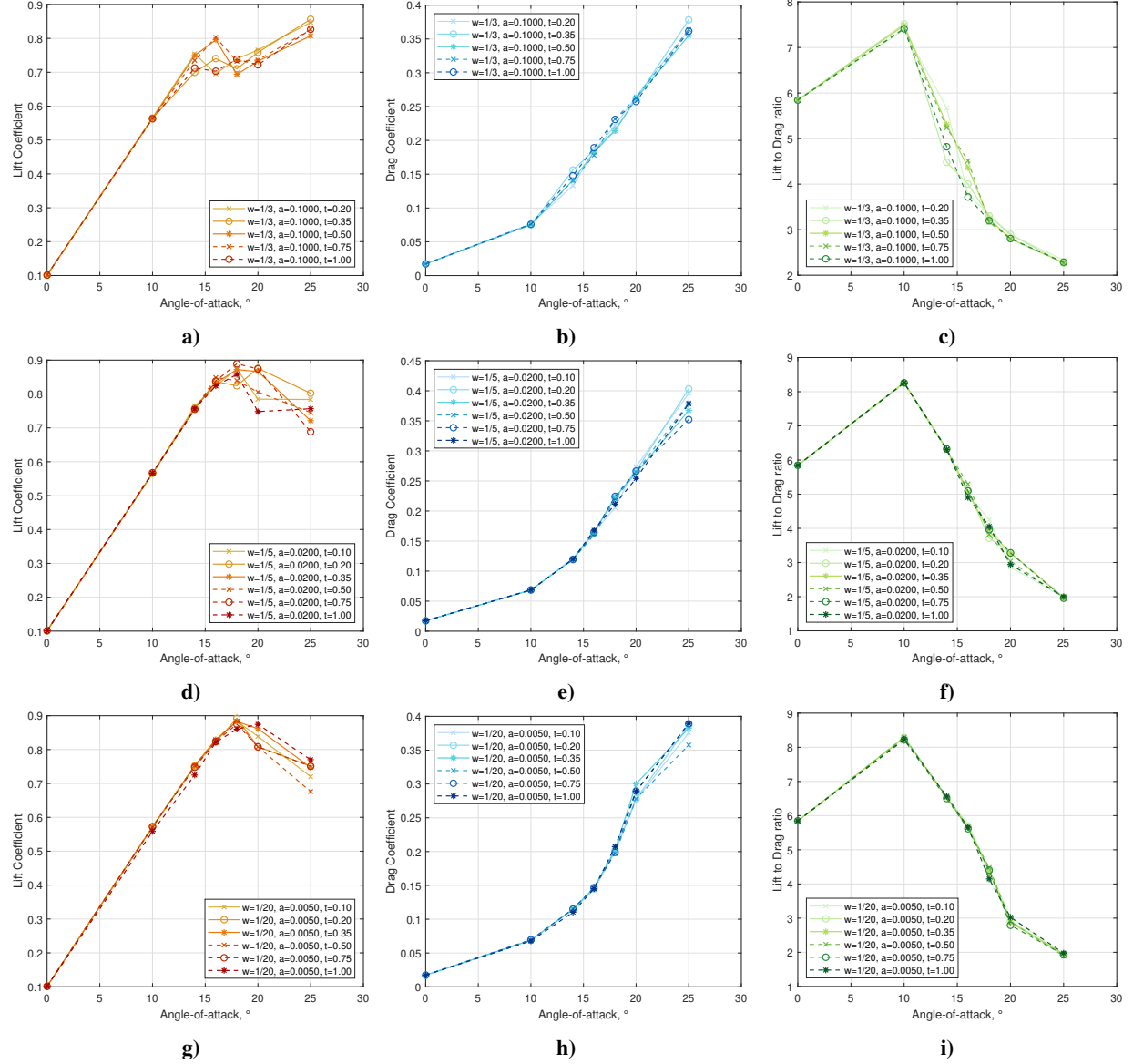


Fig. 8 Plots of the numerical data of $[1/3, 0.1, -]_{SC}$ for varying termination distances of 0.2, 0.35, 0.5, 0.75, and 1.0 for a range of AoAs, for the aerodynamic properties a) C_L , b) C_D , and c) C_L/C_D , of $[1/5, 0.02, -]_{SC}$ for varying termination distances of 0.1, 0.2, 0.35, 0.5, 0.75, and 1.0 for a range of AoAs, for the aerodynamic properties d) C_L , e) C_D , and f) C_L/C_D , and of $[1/20, 0.005, -]_{SC}$ for varying termination distances of 0.1, 0.2, 0.35, 0.5, 0.75, and 1.0 for a range of AoAs, for the aerodynamic properties g) C_L , h) C_D , and i) C_L/C_D .)

2. Termination Distance

The final geometric property that is parametrised is the termination distance of the serration dimensioned in Fig. 2. The main perturbing surface of the LE-serrations is at the LE of the wing, but the termination distance modulates how far the serration channels go back along the chord of the wing. This could affect performance and, most importantly, affect the cost of manufacturing the wings. For small values of t , the section of the wing affected by the serration is minimal, thereby minimising manufacturing costs.

The data for high-amplitude, high-wavelength serration combination $[1/3, 0.1, -]_{SC}$ are seen in Fig. 8a), 8b), and 8c) for varying values of termination distance. The most obvious result is that the termination distance is much less sensitive to aerodynamic performance. The main region of effect of the termination distance on the performance of the wing section is around stall. The low termination distances $t = 0.2$ and 0.35 seem to favour the post-stall regime, with

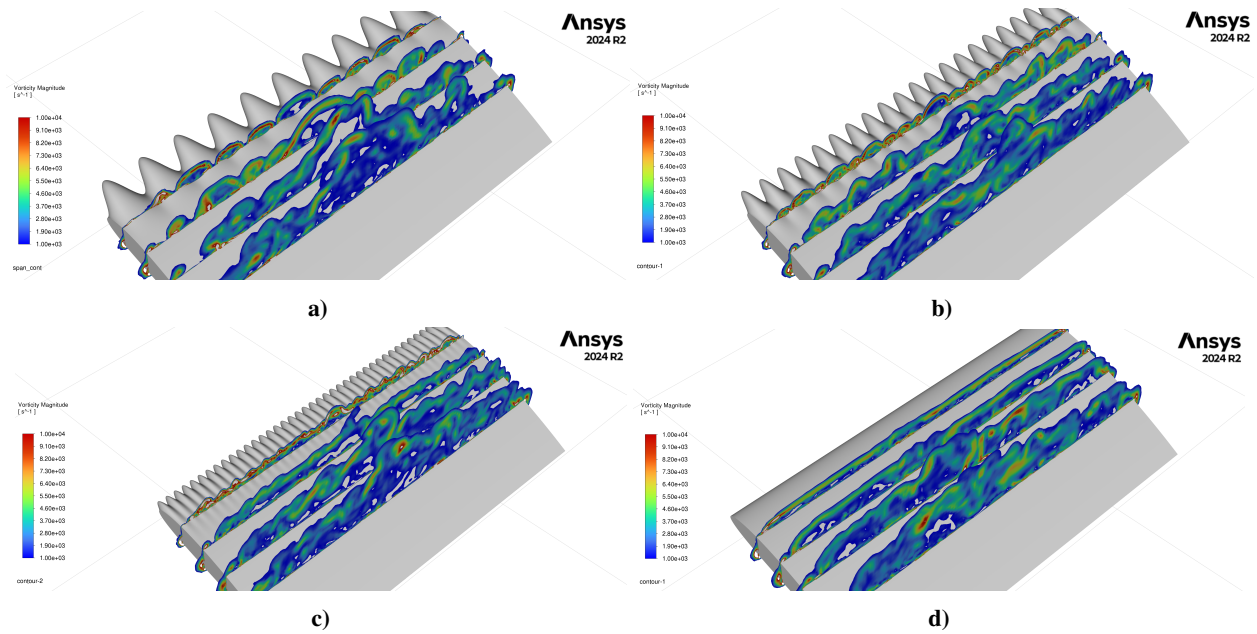


Fig. 9 Vorticity magnitude plots of planar surfaces 15 mm, 30 mm, 45 mm, and 60 mm from the LE at 25° for serration combinations of a) $[1/5, 0.075, 0.35]_{SC}$, b) $[1/10, 0.05, 0.35]_{SC}$, c) $[1/20, 0.02, 0.35]_{SC}$, and d) unserrated.

values maximising at 25° and a reduction in stall and pre-stall values. This result is also observed in Fig. 8d), 8e), and 8f), which explores a medium fidelity serration combination of $[1/5, 0.02, -]_{SC}$. The lower termination distance improves post-stall performance with minimal reduction in pre-stall performance, except at the lowest value of 0.1, which shows significant pre-stall performance reductions. This is intuitive: the shorter the termination distance, the less smooth the transition between the serrated and unperturbed sections of the wing. Whereas lower termination distances are cheaper to manufacture and favour post-stall applications, terminations that are too small result in significant performance reductions.

The least difference in performance across termination distance is that of the high-fidelity serration combination $[1/20, 0.005, -]_{SC}$ seen in Fig. 8g), 8h), and 8i). There is a small performance delta in the post-stall regime for lift, suggesting that the smaller the serration perturbations, the less the serration insertion affects performance. Whereas the performance of LE-serrated wings is much less sensitive to the termination distance of the serrations, minor optimisations must be considered when deciding how little the serration perturbs the wing through its termination distance, t .

C. Vortex Qualification

Although quantitative aerodynamic data are indicative of LE-serration effectiveness, understanding the phenomena underlying the variable effect of serration properties is important. In Fig. 9, the vorticity flow characteristics vary wildly with serration geometry. In Fig. 9a) for the large-serration combination $[1/5, 0.075, 0.35]_{SC}$, the stable vortices are held distinctly to 30% of the chord with minor vortex channels observed 45% of the chord. Looking at the smaller serrations of $[1/10, 0.05, 0.35]_{SC}$ in Fig. 9b), the vortex structures are evident at 15% chord, they are minimally present at 30% chord and nonexistent past that. This is worsened when looking at the small serrations of $[1/20, 0.02, 0.35]_{SC}$ in Fig. 9c), where there is no sign of any vortex structures past 15% chord, resembling the flow structure of the unserrated wing-section seen in Fig. 9d). The serration combination in Fig. 9a) is part of the high-performing "island" of the post-stall regime parameter optimisation seen in Fig. 7i), and the persistence of the vortex structures past 50% of the chord is a promising reason why. The total lack of stable structures in the higher-fidelity serrations is correlated with their reduction in performance post-stall. It presents a strong argument for the causation of LE-serration performance and the generation and maintenance of stream-wise vortices in a wing-section's aerodynamic performance.

V. Conclusion

This report successfully identifies the variable effect of the geometric properties of LE-serrations on the aerodynamics of wing sections. The pre-stall, stall, and post-stall regimes were each explored, and the optimal configurations were identified. The parameter space regions and trends of optimal performance result in the most general optimal configuration for all regimes of $[1/6.5, 0.03, 0.35]_{SC}$. A strong correlation between the ratio $w/a \approx 5.5$ and aerodynamic performance is observed across all stall regimes. The most evident trend is that, in lower-AoA regimes, less flow-perturbing serrations perform best, whereas greater perturbing serrations perform best at high-AoA regimes, with the generation and maintenance of streamwise stable vortex structures. The termination distance of the serrations has a minimal effect on aerodynamics, except for small insertions, which sharpen the transition from the serrations to the body and reduce performance. A value of $t = 0.2 - 0.35$ is recommended as a starting point to then optimise for slight aerodynamic metrics and manufacturing costs. The recommended future work on LE-serration properties includes exploring higher-amplitude, higher-wavelength configurations, different Re , and different airfoil profiles for different aircraft applications.

For aircraft that wish to explore the use of LE-serrations, the results from this report are auspicious and serve as a fantastic starting point for a specific case investigation in industry.

Acknowledgments

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